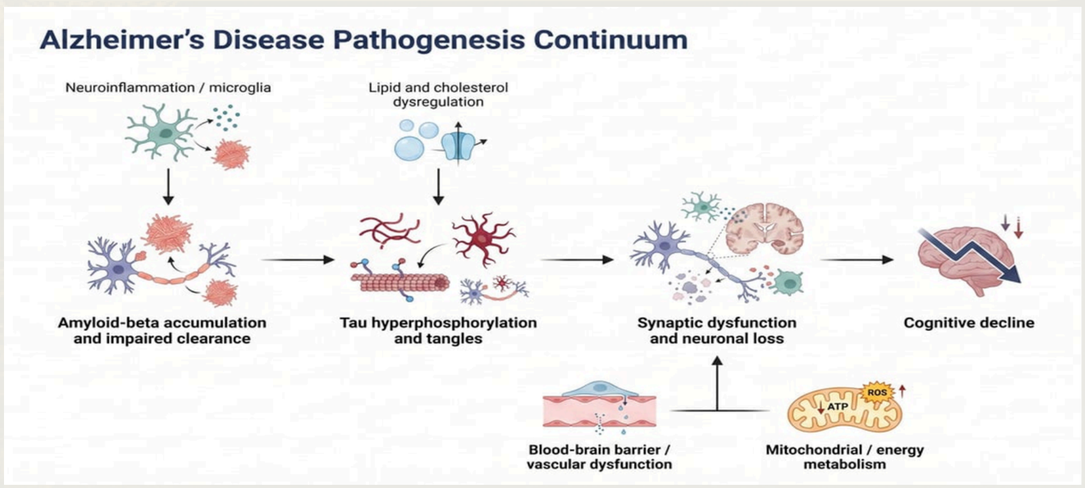
# ORCA

## On-target RNA Correction of APOE4 for Alzheimer's Disease

Can we correct disease-associated messages before they become proteins?

## 01 The Challenge

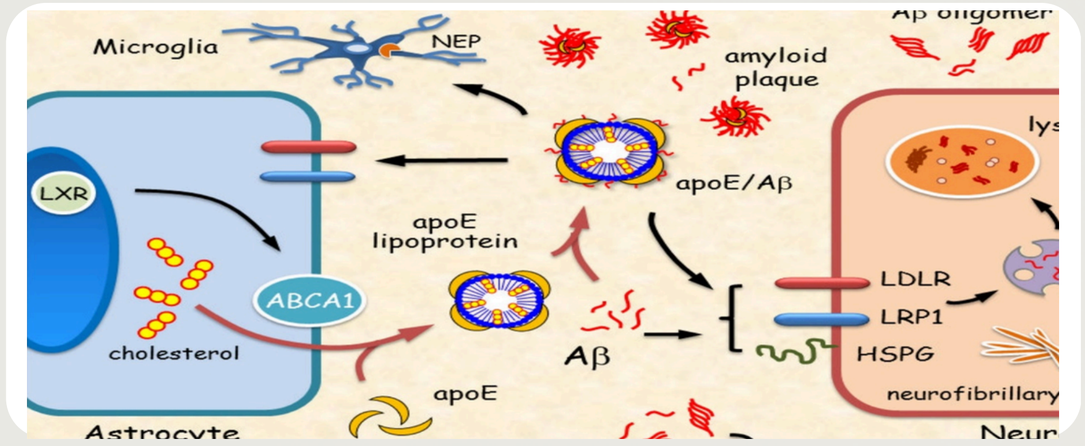
Alzheimer's disease (AD) is a major global health challenge affecting millions of people and families worldwide. Current therapies can provide symptomatic relief or target downstream pathological processes, but they do not directly address genetic risk factors such as APOE4.




## 02 Why APOE4?

APOE4 is one of the strongest genetic risk factors for late-onset Alzheimer's disease and exhibits a gene-dose effect. While anti-Aβ therapies target downstream pathology, ORCA targets APOE4-associated molecular information upstream at the RNA level.

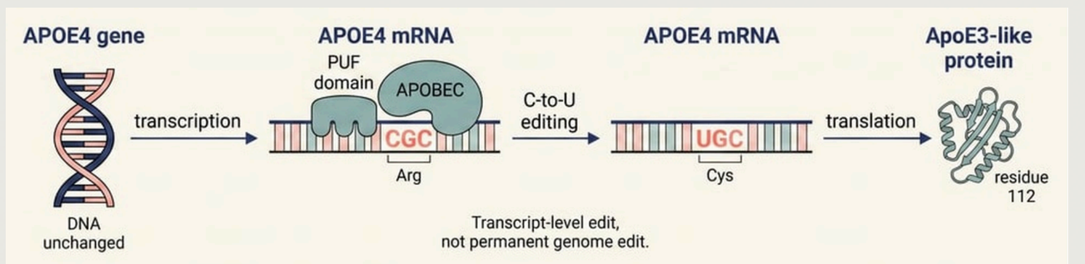
**Current anti-Aβ therapy → Targets downstream pathology**
**ORCA → Targets APOE4 RNA**



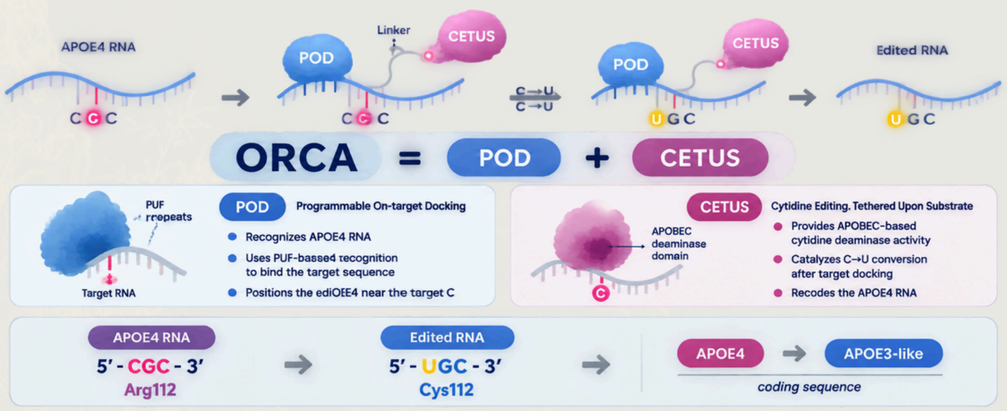

**APOE4-associated effects include:**

- Impaired lipid transport
- Reduced Aβ clearance
- Glial imbalance
- Increased neuronal vulnerability

## 03 Why RNA Editing?




We intervene at the RNA level—after transcription but before protein production. Unlike permanent DNA modification, RNA editing provides a transient intervention that can be lost as edited transcripts turn over.



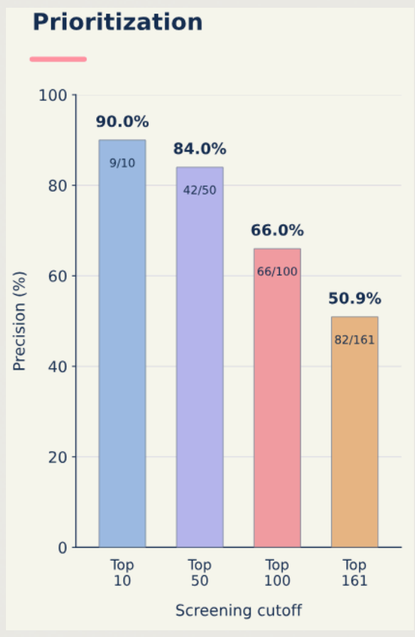

**Three design principles:**

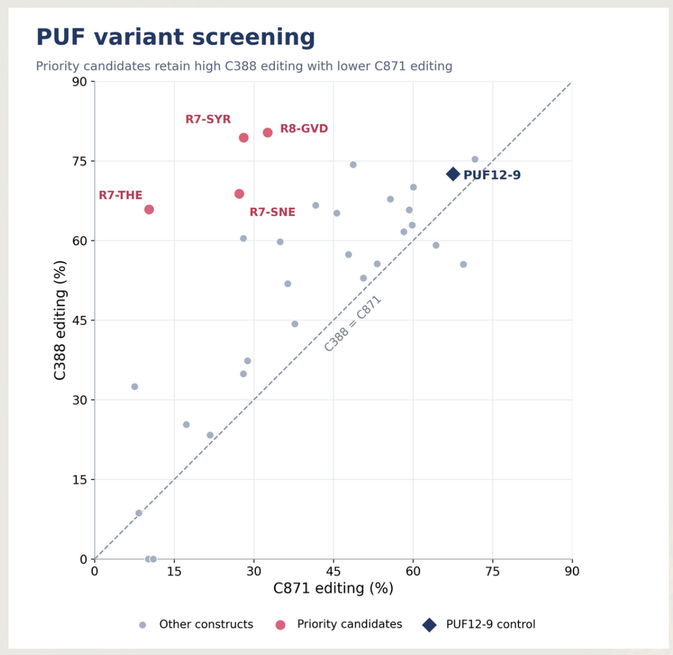
**Targeted:** Designed to recognize specific APOE4 RNA sequences.
**Transient:** Acts at the RNA level rather than permanently modifying DNA.
**Controllable:** Incorporates multiple layers for cell-specific expression, reversible suppression, and system shutdown.

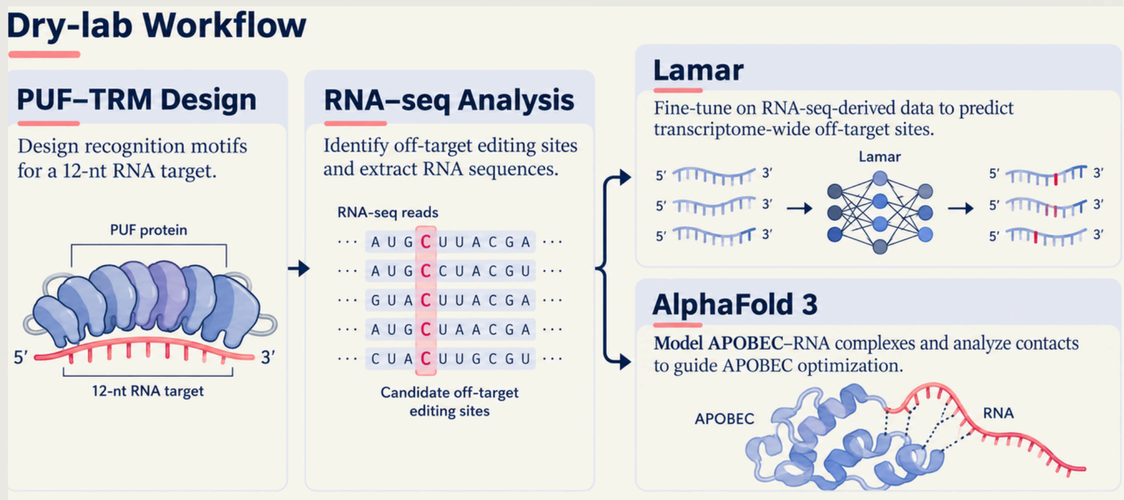
## 04 ORCA Workflow

ORCA integrates five modular stages: design, delivery, editing, control, and measurement. The system begins by defining the APOE4 target C and using LAMAR to optimize RNA recognition and minimize off-target risks. Candidate PUF-based recognition domains, linkers, and APOBEC editors are engineered to generate functional ORCA designs. The optimized cassette is delivered through an AAV-based platform with cell-specific expression control.

Upon activation, POD recognizes and docks onto APOE4 RNA, positioning CETUS to catalyze C→U editing and convert CGC to UGC, recoding Arg112 to Cys112 and generating an APOE3-like coding sequence. Multiple safety layers, including GfaABC1D, 4×6T MRE, ASO, and Cre–loxP, provide expression restriction, reversible suppression, and system shutdown. ORCA performance is validated through molecular, expression, and functional assays to evaluate editing efficiency, specificity, and biological outcomes.




## 05 Therapeutic Positioning

| Time / Stage | Therapy | Therapeutic focus |
|---|---|---|
| 1990s–2000s | Donepezil / Memantine | Symptomatic relief |
| 2023--2024 | Lecanemab / Donanemab | Amyloid-targeted therapy |
| ORCA | Targeted RNA editing of APOE4 | APOE4-associated molecular intervention |

Rather than replacing existing approaches, ORCA is being developed as a potential upstream strategy to modulate APOE4-associated molecular risk.

## 06 Potential Impact

Although ORCA is designed around APOE4 and Alzheimer's disease, its modular architecture may provide a framework for future RNA-editing research.

- **Neurodegenerative disease research**
- **Precision medicine**
- **Programmable RNA editing**
- **Modular therapeutic design**

**Programmable. Targeted.**
**RNA-level.**
**Editing RNA, not DNA.**

